SIPF 2026

The many times of action observation on corticospinal excitability: Coupling online facilitation with after-effects.

Ciraolo A., Bazzini M.C., Nuara A., Avanzini P., Fabbri-Destro M.
Department of Medicine and Surgery, University of Parma

Does observation leave a trace?

Online motor resonance is well established. Does it leave a measurable corticospinal trace after observation?

We tested whether individual AO responsiveness predicts subsequent resting excitability.

Experimental design

32 healthy right-handed adults
Within-subject design

BASELINE
Rest

⬇

ACTION OBSERVATION
VR · TMS during observation
Ego 2D Ego 3D
Allo 2D Allo 3D

⬇

POST-SESSION
Rest

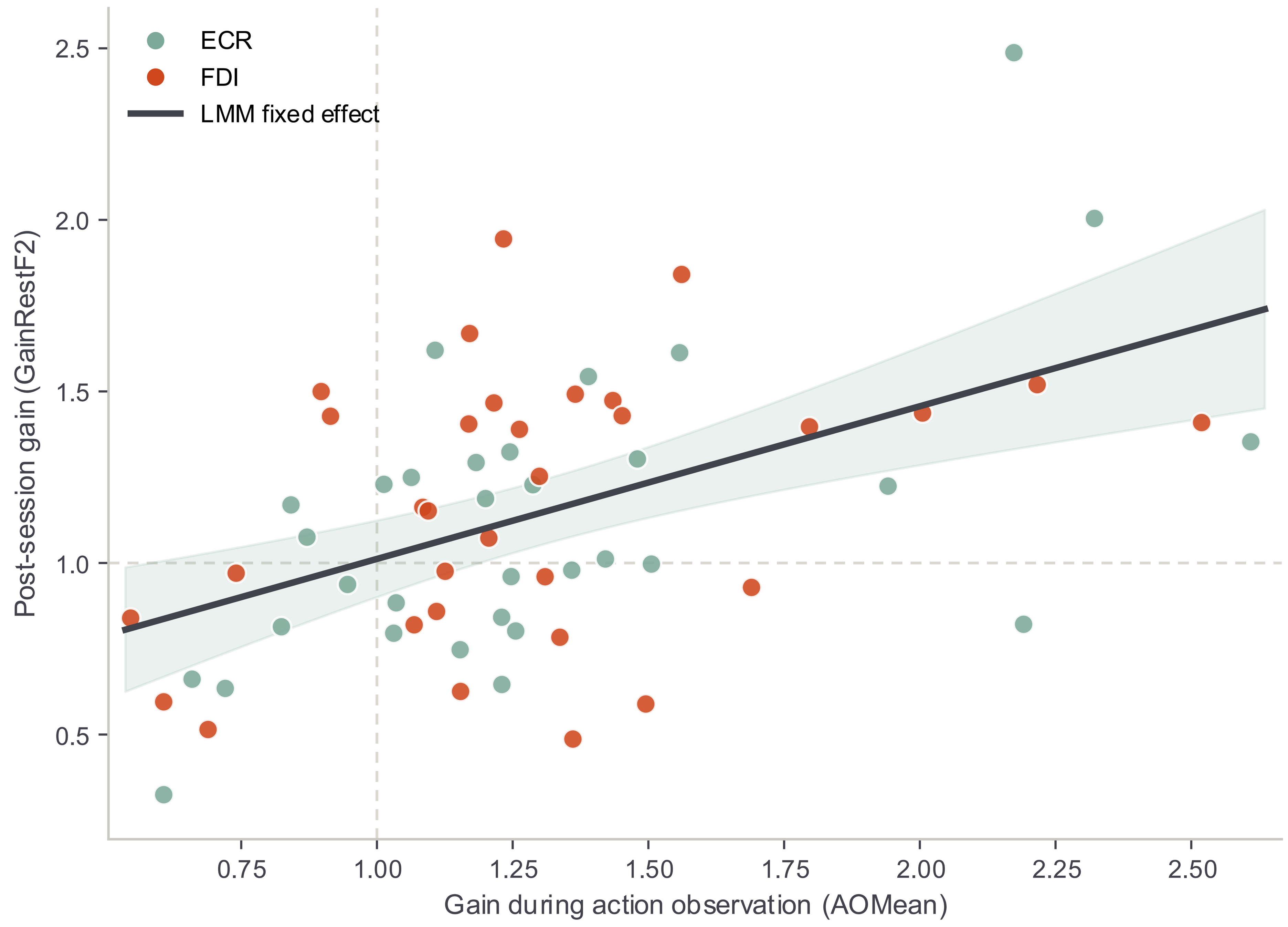
TMS: left M1 EMG: ECR / FDI

ONLINE FACILITATION → AFTER-EFFECT
Both indices: MEP gain relative to baseline rest

Stronger online facilitation predicts stronger post-session corticospinal excitability

β = 0.445 | **p < .001** | **R²m = .247**

Participants showing stronger facilitation during AO also showed stronger post-session excitability.

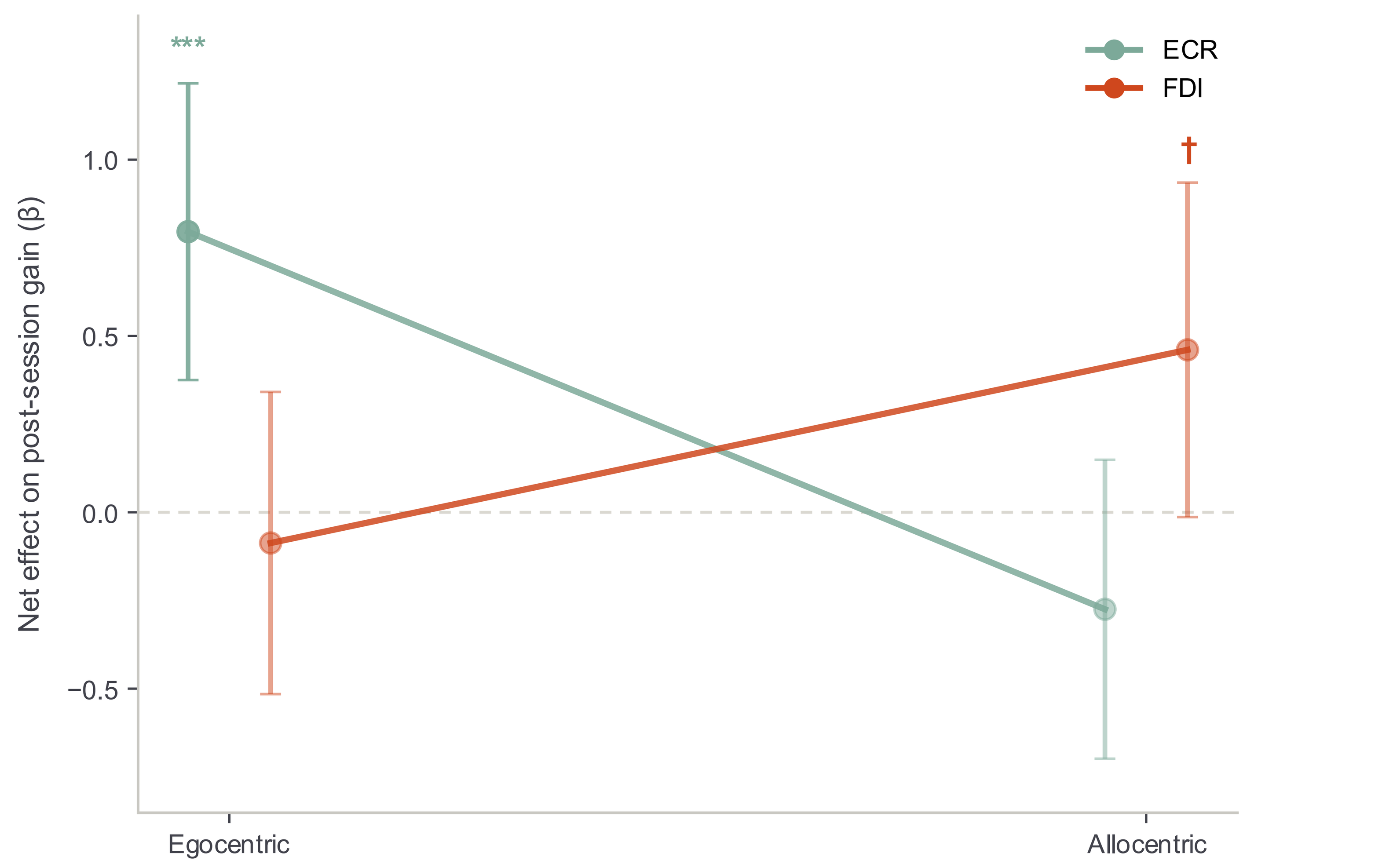


Action observation reliably facilitates CSE

Facilitation was consistent across viewing conditions, with no detectable modulation by perspective or dimensionality.

The online–after relationship shows perspective–muscle specificity

Perspective × Muscle p = .014



Egocentric → ECR
β = .796 p < .001

Allocentric → FDI
β = .461 p = .057 (trend)

Motor resonance does not end when observation stops.

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Perspective–muscle specificity may be an additional factor to consider when designing AOT protocols.

Naish KR, Houston-Price C, Bremner AJ, Holmes NP. (2014). Neuropsychologia, 64, 331–348.

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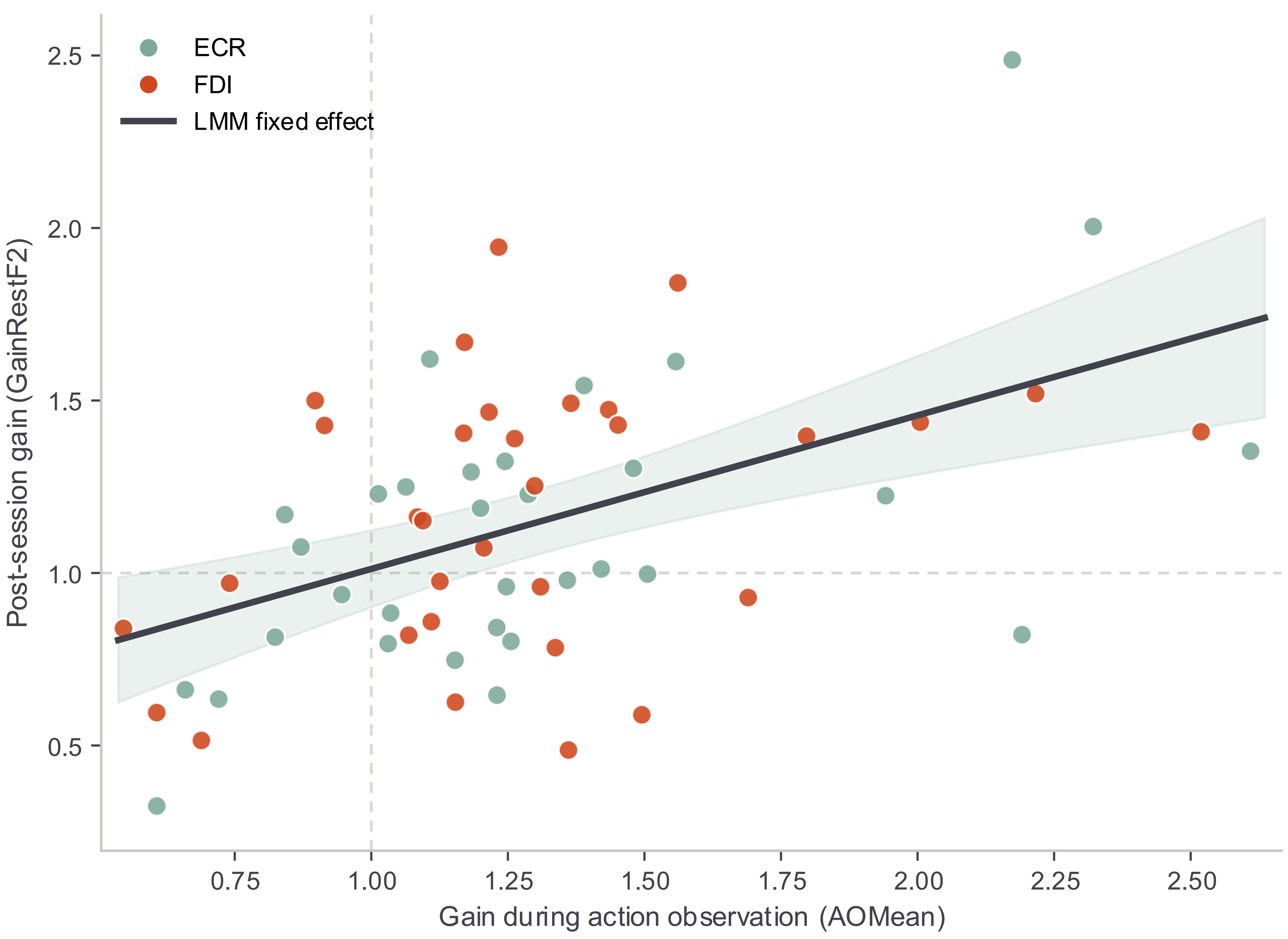

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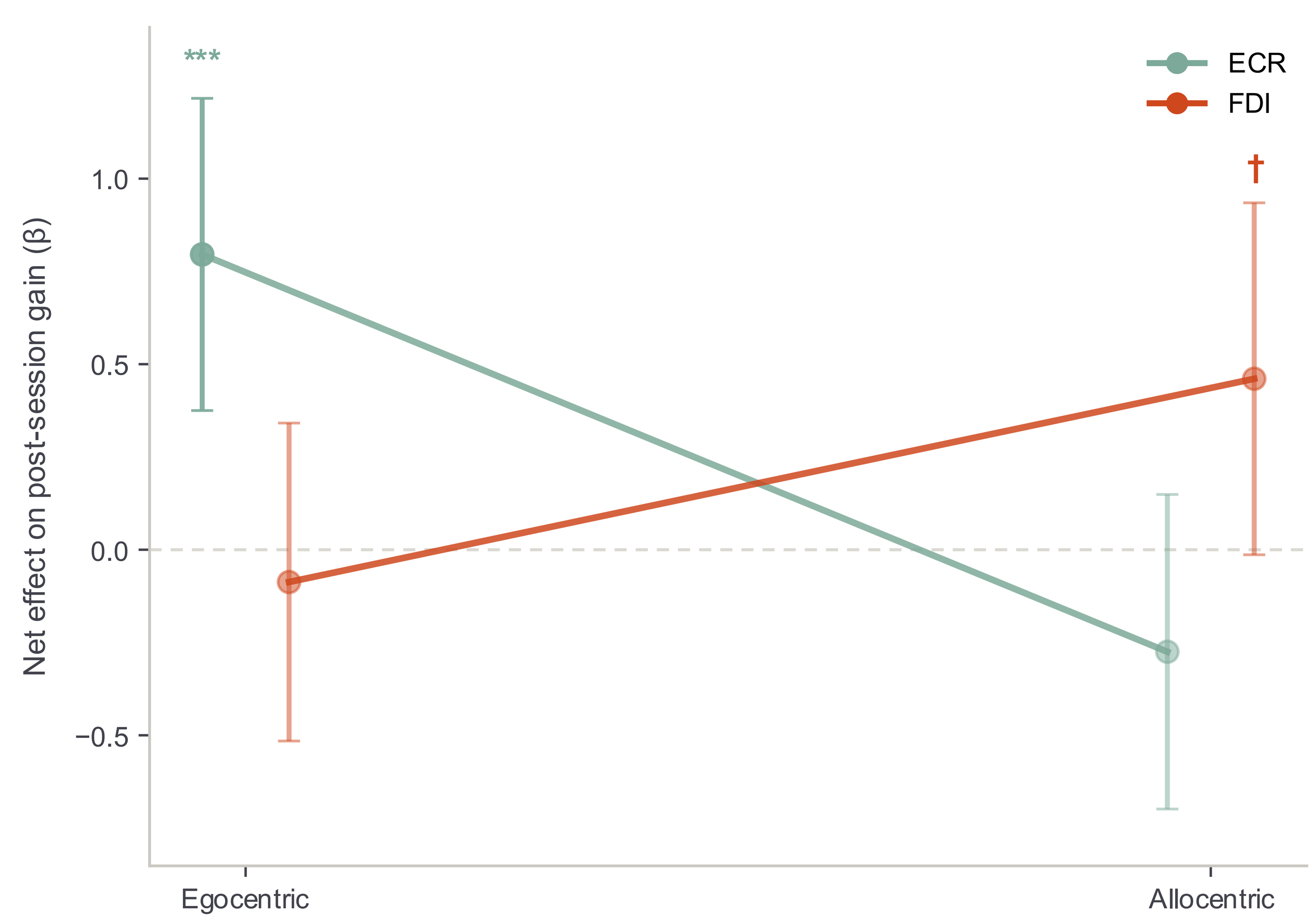

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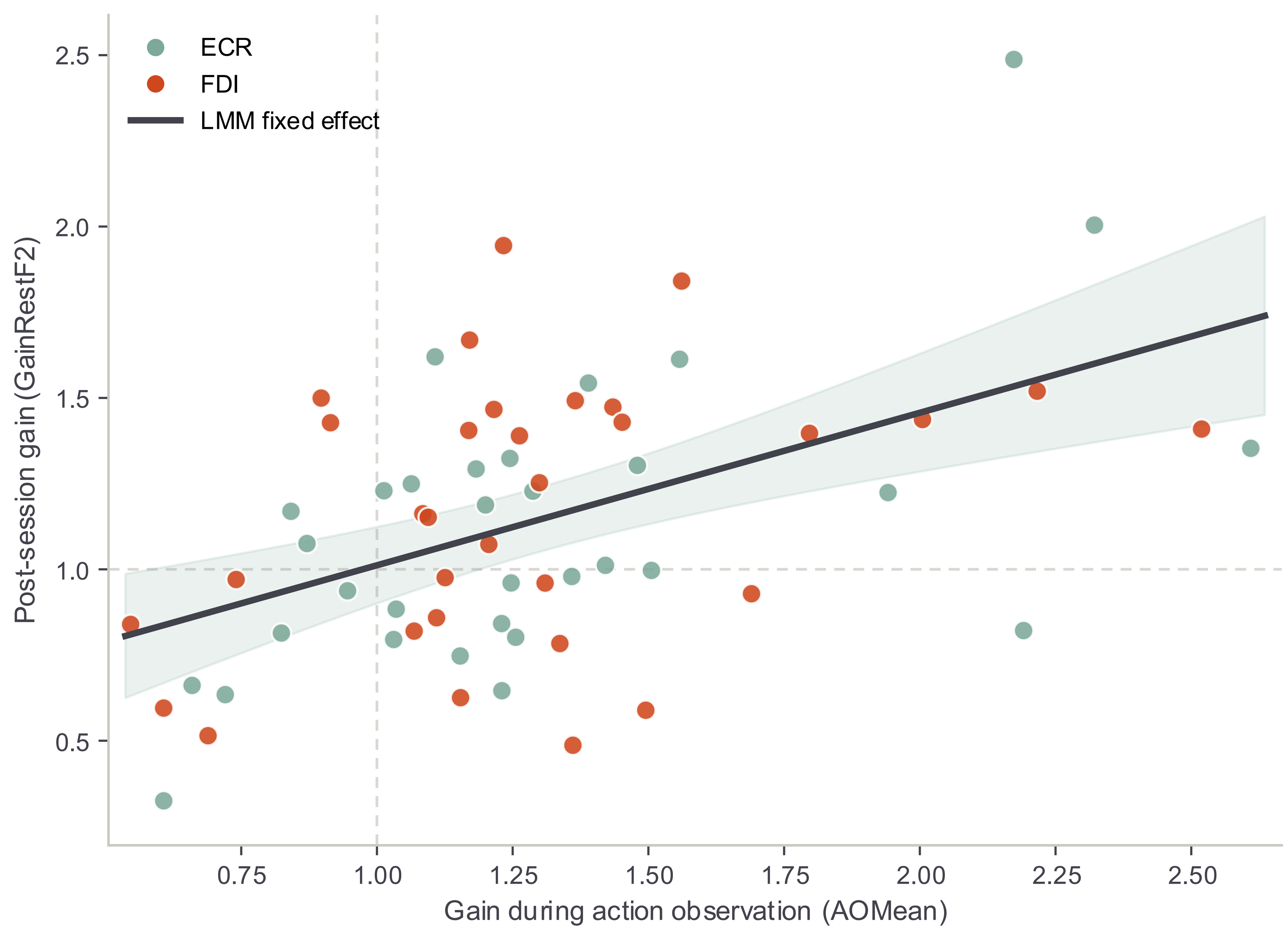

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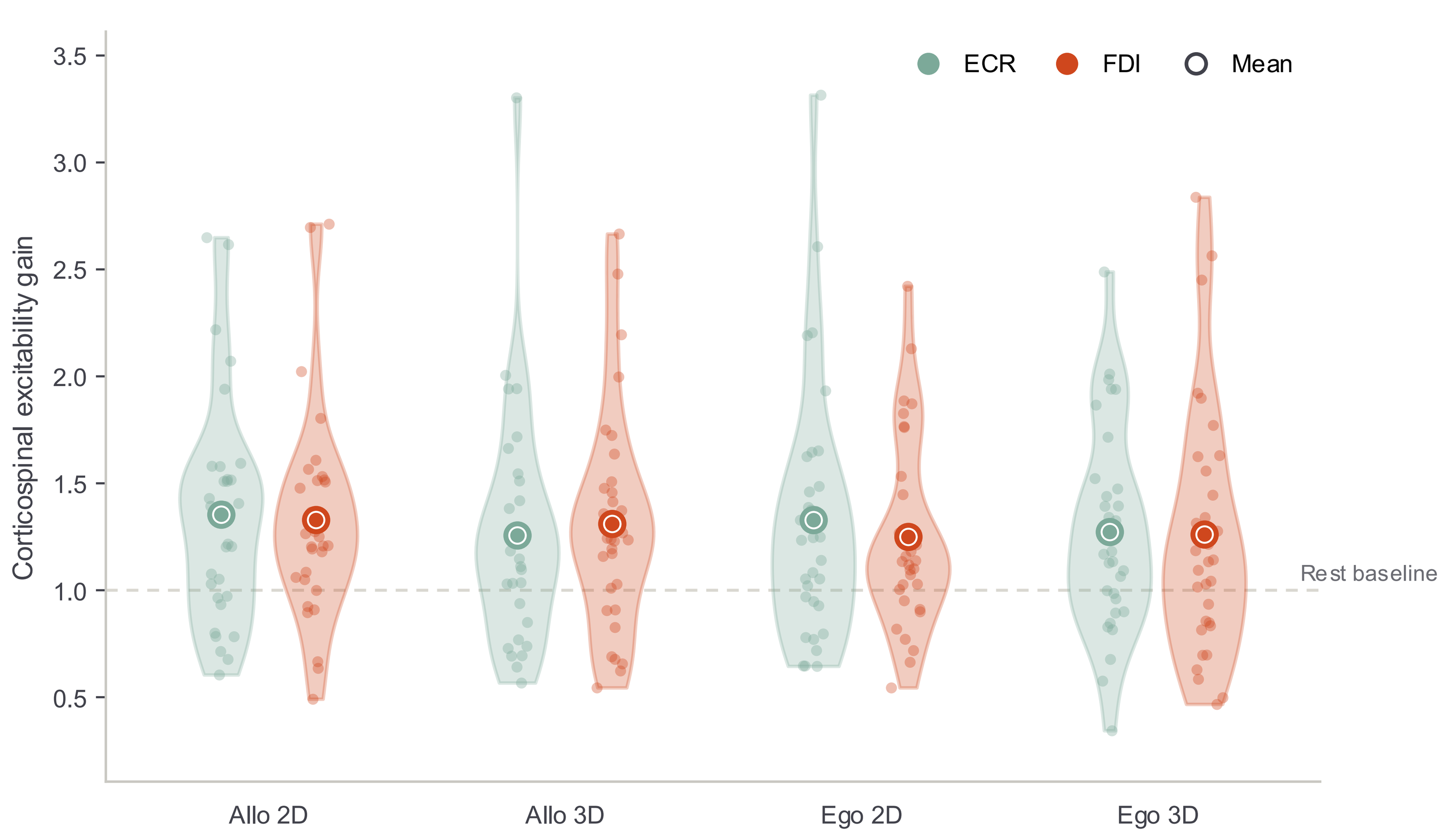

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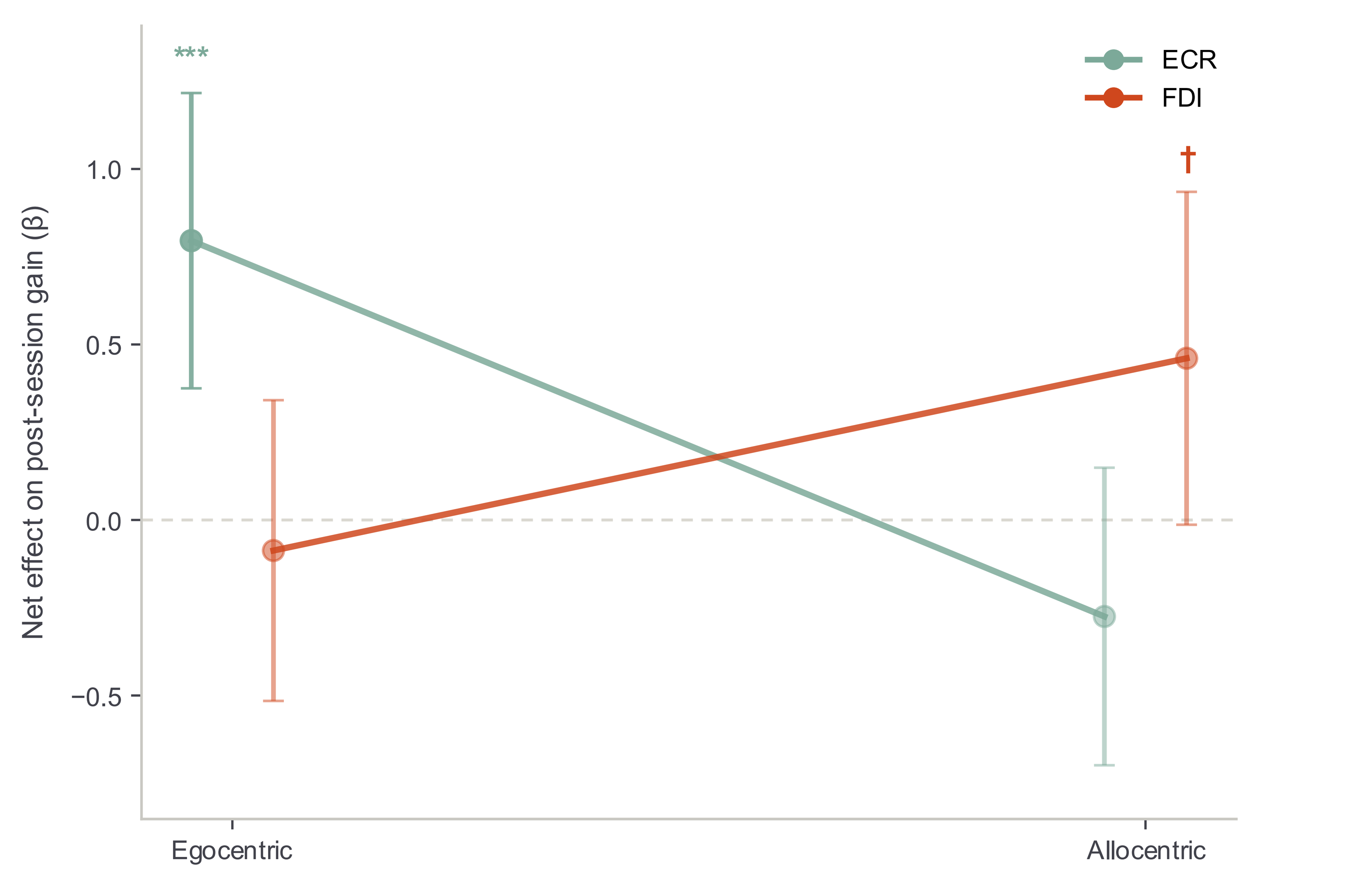

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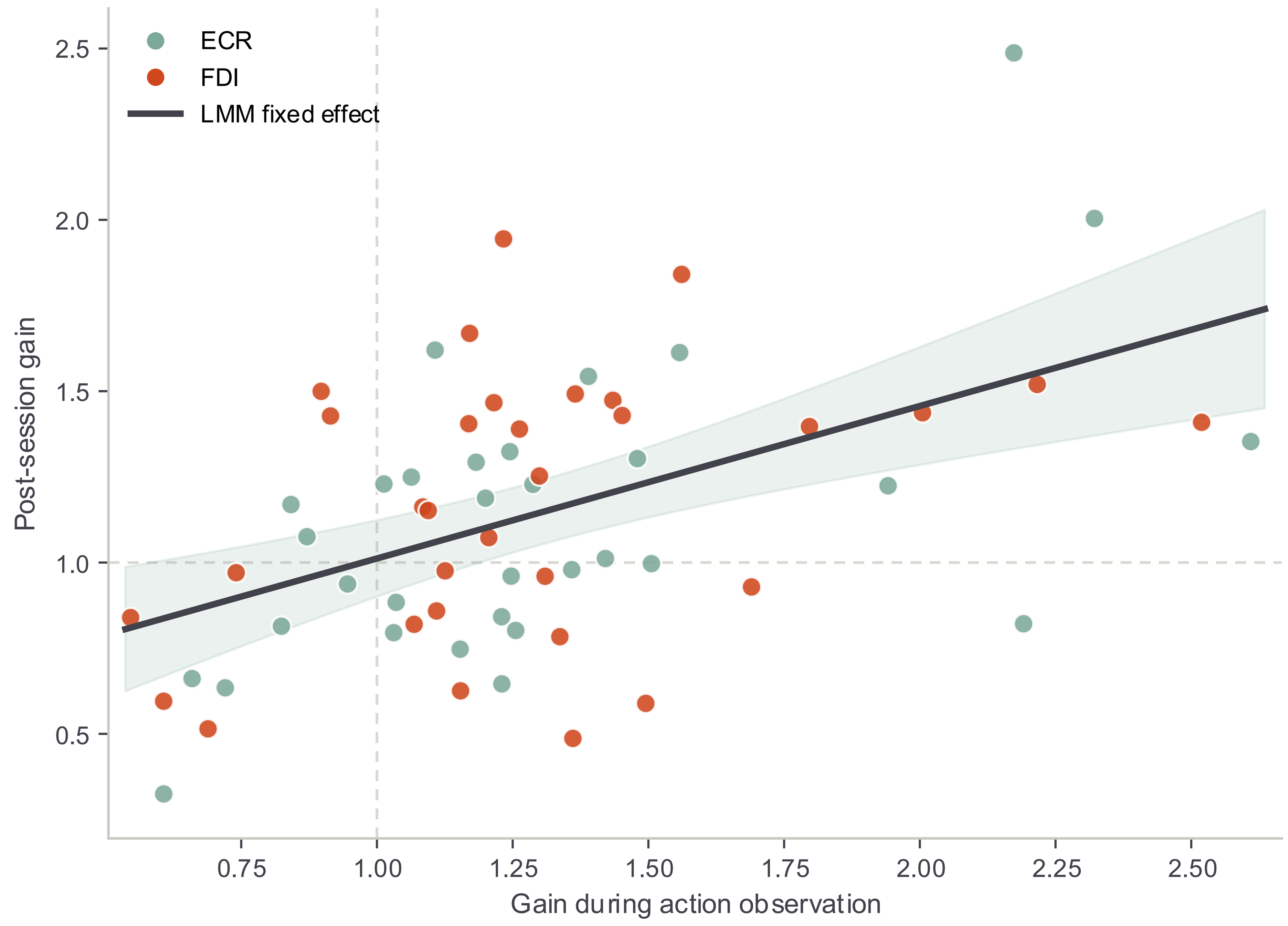
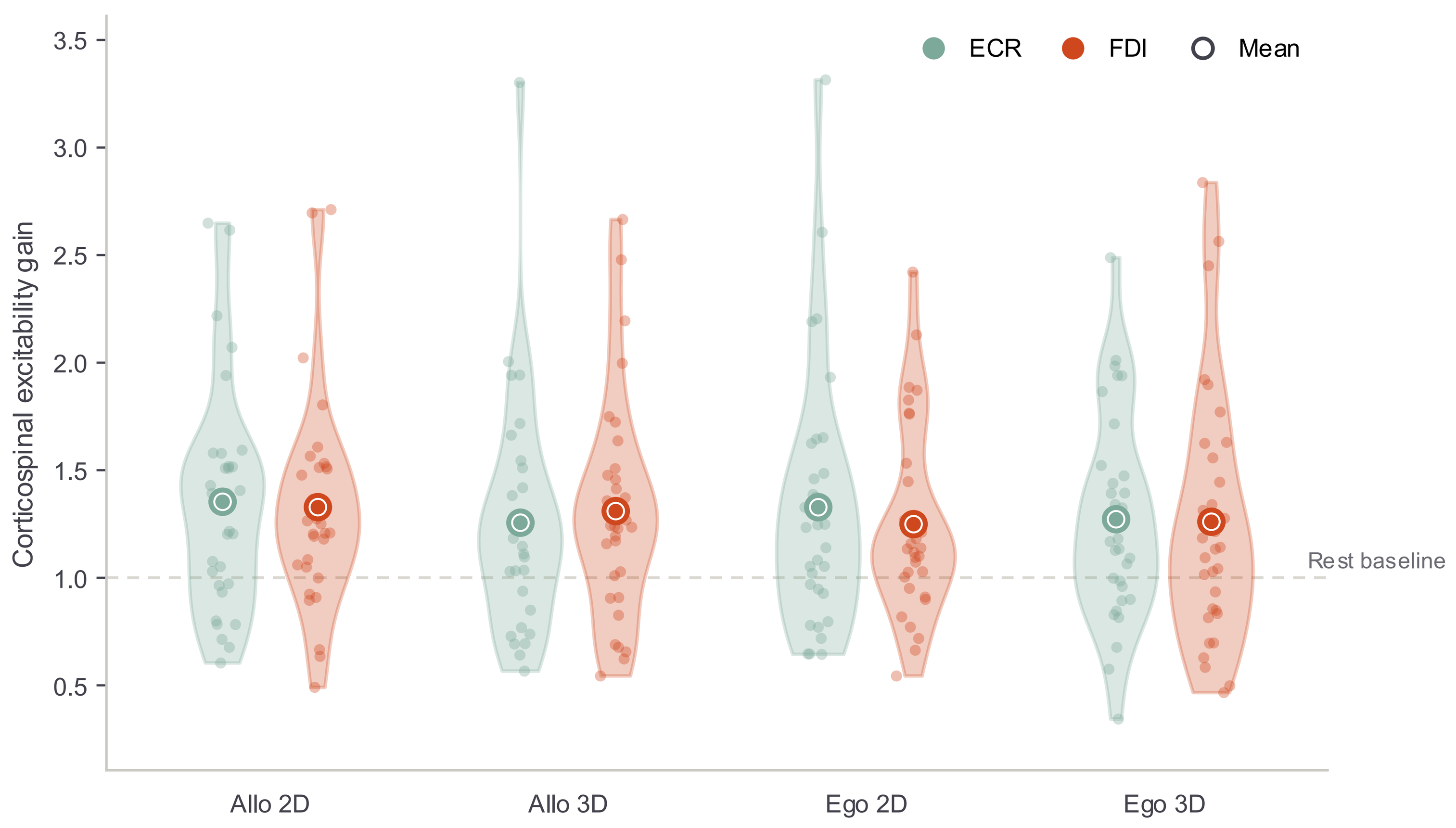


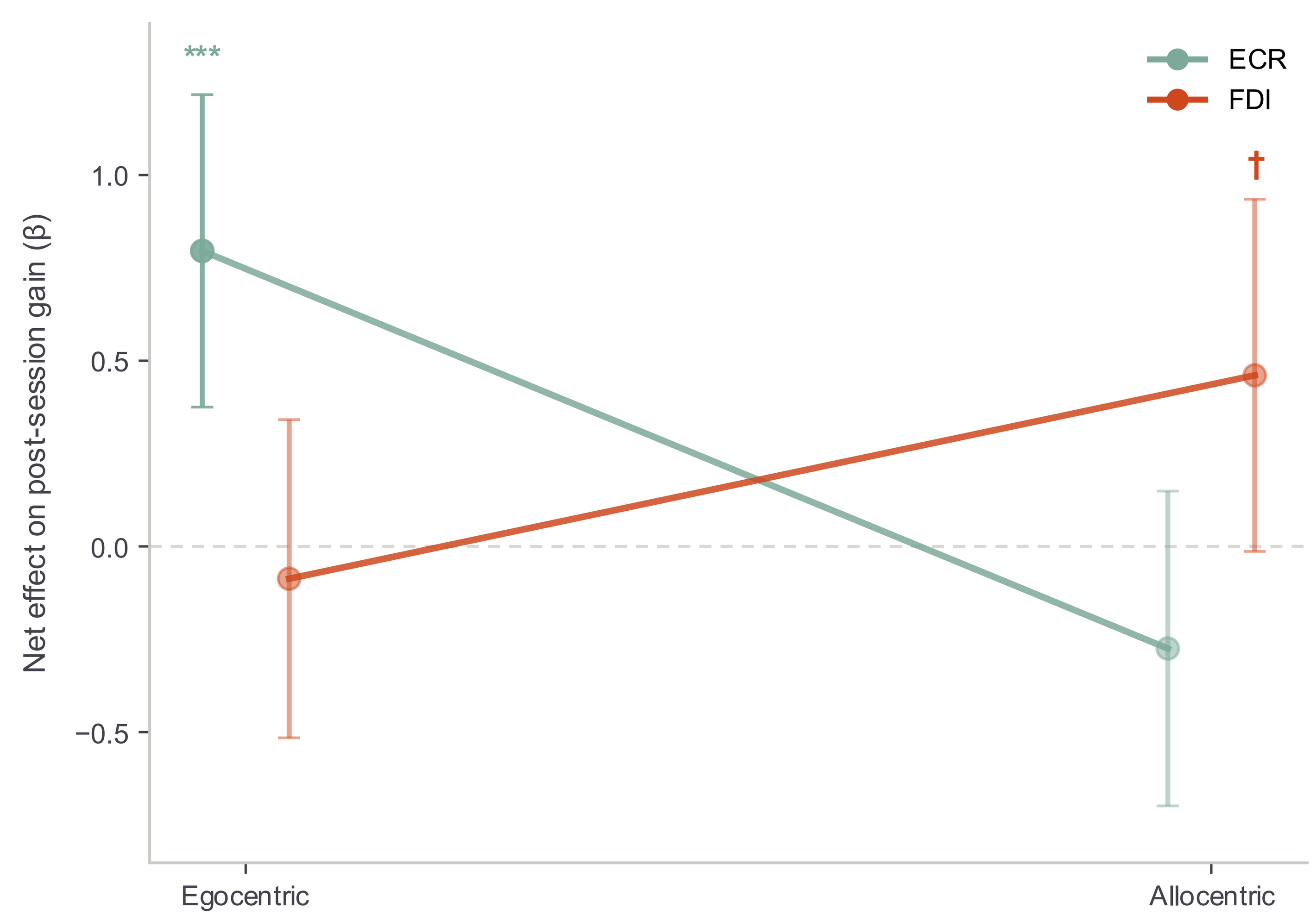
Action observation reliably facilitates Corticospinal Excitability



Facilitation was consistent across the four viewing conditions.

≈1.33× baseline
mean corticospinal excitability during AO
No differences across viewing conditions
ECR p = .322 · FDI p = .409

The online–after relationship shows perspective–muscle specificity



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Perspective–muscle specificity may be an additional factor to consider when designing Action Observation Treatment protocols.

Next step: a sham-TMS control experiment using the same visual conditions is planned to disentangle AO-driven after-effects from potential TMS-related contributions.

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